IBRAHIM MUSTAFA OPU

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SUMMARY

Self-taught programmer in Toronto, on a gap year before starting a computer science degree. I designed and published Whisper, a programming language with conversational English syntax, interpreted in Python and available on PyPI with a VS Code extension. I write mostly Python and JavaScript, and I am working through data structures and algorithms, Harvard's CS50x and discrete math, all in public on GitHub. Looking for junior or entry-level development work.

PROJECTS

Whisper — programming language

Python · Interpreter

github.com/ibrahim787898m/whisper-lang · pypi.org/project/whisper-lang · whisper.ibrahimmustafaopu.com

- Designed and built an interpreted language with conversational English syntax: a line-based interpreter in Python that matches English-style statements, runs indented blocks and user-defined functions, and evaluates expressions in a restricted scope.
- Published to PyPI as whisper-lang (Python 3.7+, no dependencies, MIT), with a companion VS Code extension for syntax highlighting and a documentation site.
- Implemented story objects, a feature for declaring characters and items in plain sentences; resolved attribute collisions between objects by giving each its own namespace instead of a single shared lookup table.

Grocery Split — real-time web app

JavaScript · Firebase · HTML/CSS

github.com/ibrahim787898m/grocery-split · grocery-split.ibrahimmustafaopu.com

- Shared expense tracker in continuous weekly use by two people on separate devices, with real-time sync via Firebase Realtime Database, password sign-in, per-month views and automatic split calculation.
- Two clients writing at once exposed weaknesses in my first data model; it taught me why data modelling and concurrent writes matter, and they are what I am now studying properly.

Online Diary — web app

JavaScript · Firebase

github.com/ibrahim787898m/diary-webapp · diary.ibrahimmustafaopu.com

- Private, password-protected journalling app for personal use, with entry creation, sorting and organisation — my first project deployed with a real backend.

NLP toolkit — command-line tool

Python · spaCy

github.com/ibrahim787898m/NLP_SpaCy

- CLI tool running eight NLP operations on any input text — tokenisation, linguistic annotation, lemmatisation, sentence detection, POS tagging, named entity recognition, stop-word removal and dependency parsing — with most results shown as formatted tables.

LEARNING IN PUBLIC

Data structures & algorithms — notes, quizzes and LeetCode solutions in Python github.com/ibrahim787898m/dsa — Ongoing

CS50x — lecture notes and weekly reflections github.com/ibrahim787898m/cs50x — Ongoing

Discrete math — logic, proofs, sets and functions github.com/ibrahim787898m/math — Ongoing

TECHNICAL SKILLS

Confident HTML, CSS, JavaScript, Python, Git, responsive layout

Working with Firebase, pytest, data structures and algorithms, C, C++, discrete math

Next up React, TypeScript, SQL

EDUCATION & COURSES

CS50P — Introduction to Programming with Python, Harvard University certificate — 2025

CS50x — Introduction to Computer Science, Harvard University — In progress

Higher Secondary School Certificate — Sylhet Government College, Science, GPA 5.00/5.00 — 2022

Secondary School Certificate — Starlight Academy, Science, GPA 5.00/5.00 — 2020

ADDITIONAL

Languages English, Bengali, Hindi